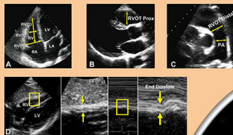




Chamber Quantification



Variable	Abnormal
A. RV Basal (RVBD)	> 4.2 cm
B. RV Mid (RVMD)	> 3.5 cm
C. RV Longitudinal (RVLD)	> 8.6 cm
D. RVOT PLAX distal	> 3.3 cm
E. RVOT Axial proximal	> 2.7 cm
F. RV Wall Thickness	> 0.5 cm

Right Atrium



Tracing of the right atrium (RA) is performed from the plane of the tricuspid annulus (TA), along the posterior septum (AS), superior and antero-lateral walls of the RA. The RA major dimension is measured from the TA center to the superior RA wall, and the RA minor dimension is measured from the antero-lateral wall to the AS, as indicated by the yellow arrows.

Variable	Abnormal
RA Major Dimension	> 5.3 cm
RA Minor Dimension	> 4.4 cm
RA End-Systolic Area	> 18 cm ²

Right Atrial Pressure

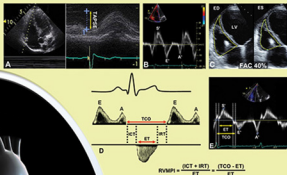
Estimation of RA pressure on the basis of IVC diameter and collapse

Variable	Normal (0-5 (3) mmHg)	Intermediate (5-10 (8) mmHg)	High (10-15 mmHg)
IVC diameter	< 2.1 cm	> 2.1 cm	> 2.1 cm
Collapse with sniff	> 50%	< 50%	> 50%

* Ranges are provided for low and intermediate categories; however, for simplicity, a mid-range value of 3 mmHg for "normal" and 8 mmHg for "intermediate" are suggested. Intermediate (8 mmHg) RA pressures may be downgraded to normal (3 mmHg) if no secondary indices of elevated RA pressure are present, upgraded to high if minimal collapse with sniff (< 50%) and secondary indices of elevated RA pressure are present, or left as 8 mmHg if uncertain.

IVC = inferior vena cava; RA = right atrium; RV = right ventricle

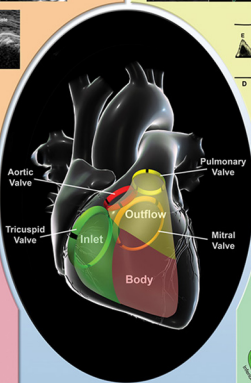
Systolic Function



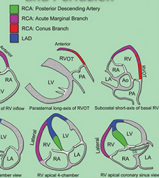
TCO = Time from closure to opening of tricuspid valve
ET = Ejection time
ED = End-diastole
ES = End-systole
RVMPI = Right ventricular myocardial performance index

Variable	Abnormal
A. TAPSE	< 1.6 cm
B. Pulsed Doppler peak velocity at the annulus (ST)	< 10 cm/s
C. FAC (%)	< 35%
D. Pulsed Doppler MPI	> 40
E. Tissue Doppler MPI	> 55

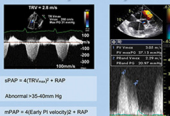
Figure D adapted from Assessment of the Right Ventricle by Echocardiography: A Primer for Cardiac Sonographers. J Am Soc Echocardiogr 2009; 22:776-792.



Segmental Analysis and Perfusion



Pulmonary Hemodynamics



$$sPAP = 4(TRV_{max})^2 \div RAP$$

Abnormal: > 35-40 mm Hg

$$mPAP = 4(Early P1 velocity)^2 \div RAP$$

$$mPAP = 1/3 sPAP + 2/3 dPAP$$

$$mPAP = 79 - 0.45 \times \text{Acceleration Time}$$

Abnormal: > 25 mm Hg

$$dPAP = 4(End P1 velocity)^2 \div RAP$$

sPAP = Systolic pulmonary artery pressure
mPAP = Mean pulmonary artery pressure
dPAP = Diastolic pulmonary artery pressure
RAP = Right atrial pressure

Diastolic Function

	E-A	E'E'	Deceleration Time	Additional Findings
Normal	0.8-2.1	< 6	> 120 ms	-
Impaired Relaxation	< 0.8	< 6	> 120 ms	-
Pseudonormal	0.8-2.1	> 6	> 120 ms	Diastolic flow predominance in RV
Restrictive	> 2.1	> 6	< 120 ms	Late diastolic antegrade flow in PA